

WASP-39b

R-1772

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_130313 · created 2026-08-02T22:15:11+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456364.8924 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1415 +/- 0.0074
Transit depth (Rp/Rs)^2	2.0 +/- 0.21 [%]
Orbital Inclination (inc)	88.12 +/- 0.74
Airmass coefficient 1 (a1)	1.157 +/- 0.012
Airmass coefficient 2 (a2)	-0.1097 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	None
Optimal Aperture	4.15
Optimal Annulus	11.51
Transit Duration (day)	0.1212 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1415 $\pm$ 0.0074 vs 0.1457 (deviation 0.51 σ)
Duration fit vs published	0.1212 d vs 0.12 d (deviation 0.17 σ)
Mid-time O-C	-11.2 min
Depth SNR	17.3
Residual scatter	1.06 %

Light curve

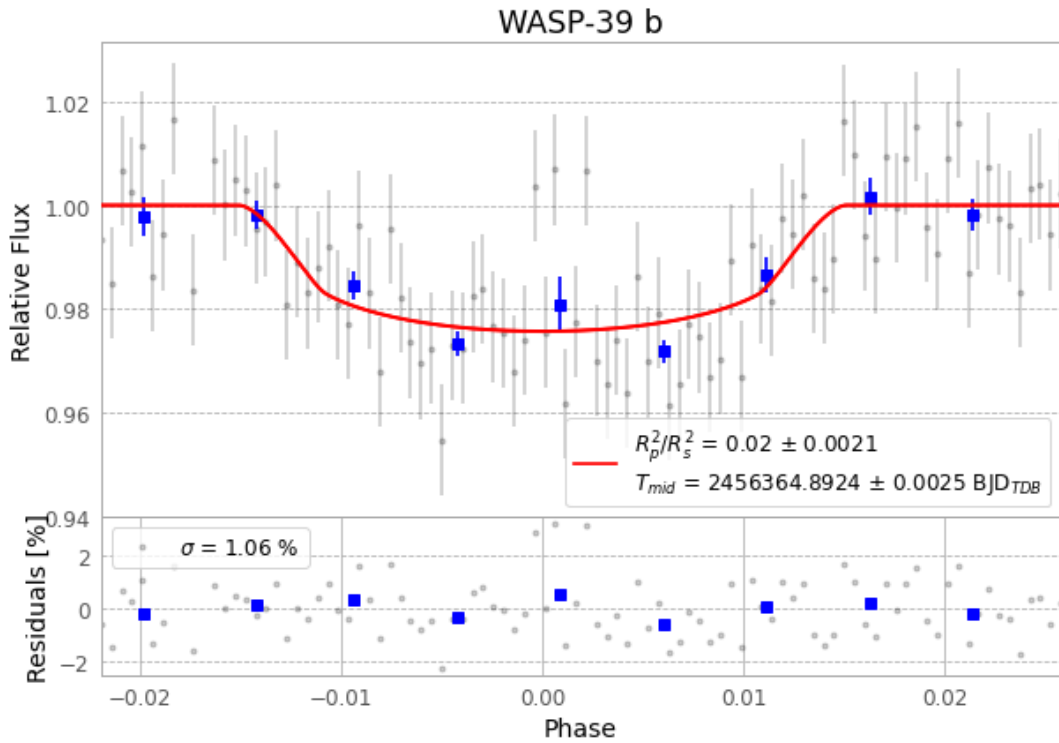


Figure 1 — FinalLightCurve_WASP-39 b_2013-03-13.png (SHA-256 941ccb8b17245f29...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7fe6f751e27e555...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), WASP-39130313070912.FITS → WASP-39130313114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-130313.log (a297ef3bf89d...), FinalLightCurve_WASP-39 b_2013-03-13.png (941ccb8b1724...), FinalLightCurve_WASP-39 b_2013-03-13.csv (7299ef13df2f...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)